

EXPLORATORY ANALYSIS OF RAINFALL DATA IN INDIA FOR AGRICULTURE

DATE	28-02-2026
TEAM ID	LTVIP2026TMIDS89121
PROJECT NAME	Exploratory Analysis of Rainfall Data in India for Agriculture
MAXIMUM MARKS	4 MARKS

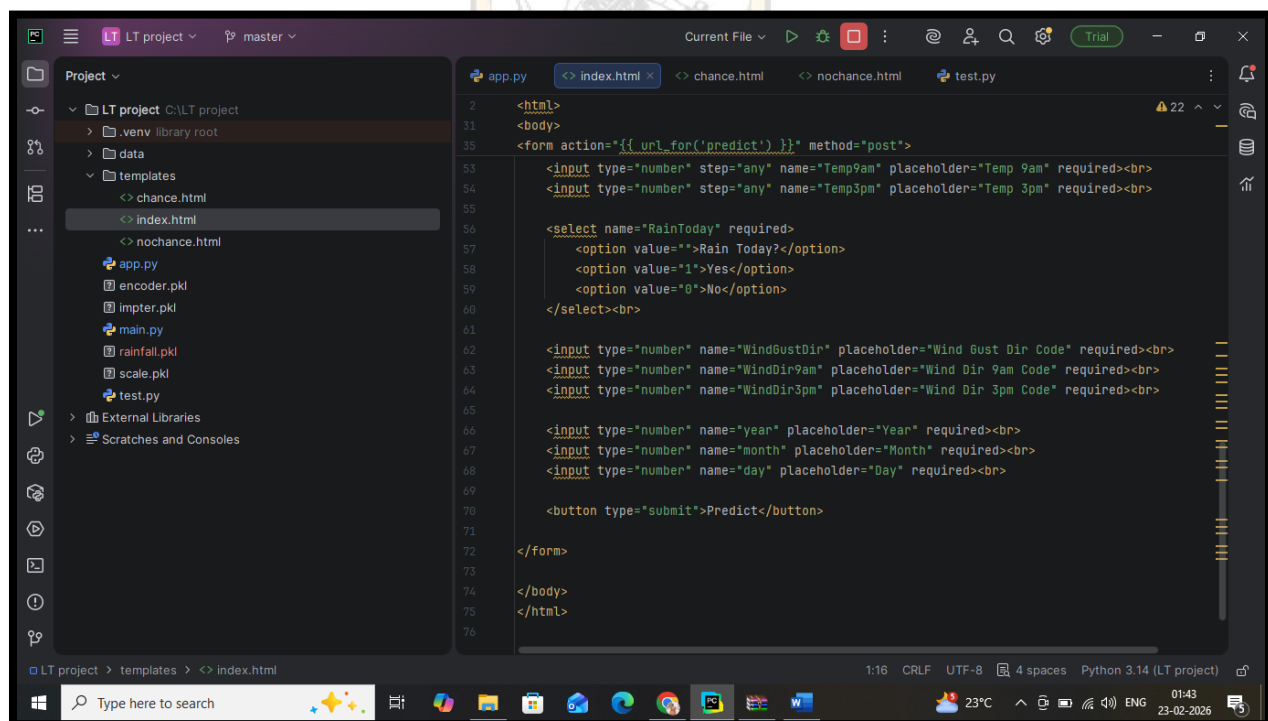
6.5 - APPLICATION BUILDING:

In our project we have 3 HTML files

1. index.html
2. chance.html
3. nochance.html

1.index.html:

- We have to paste code.
- On the html file only

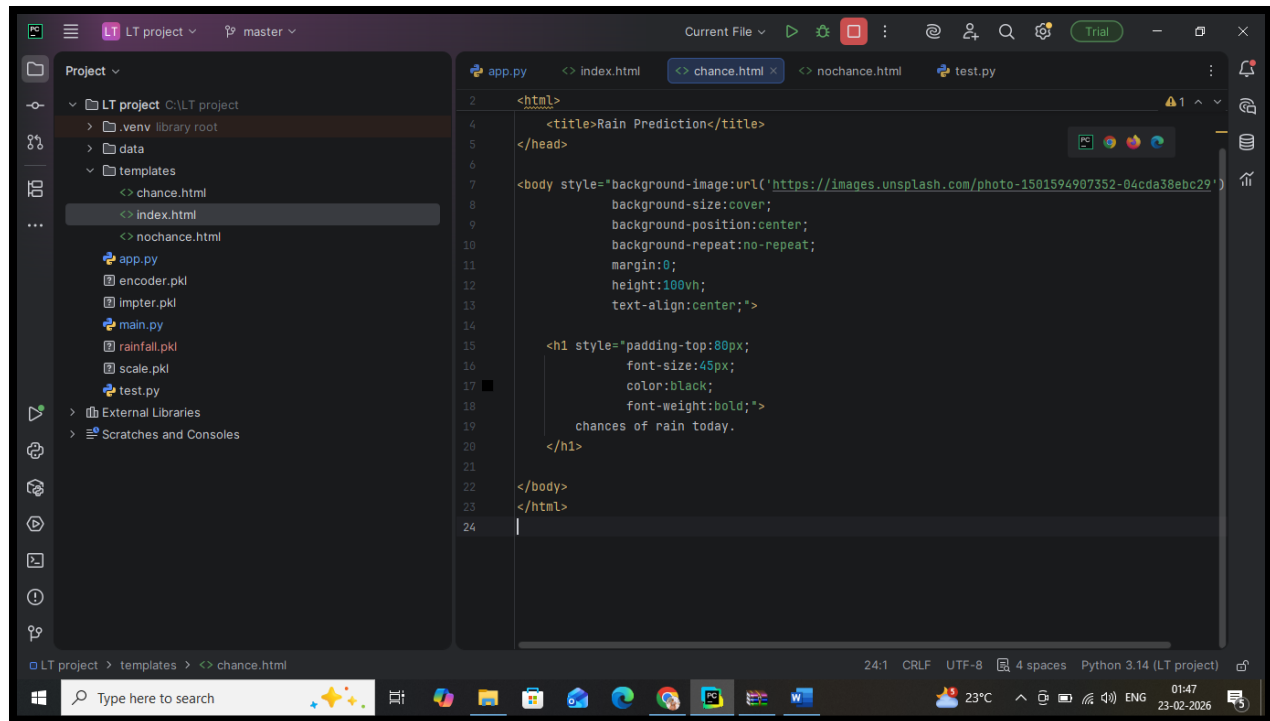


```
1 <html>
2 <body>
3   <form action="{url_for('predict')}" method="post">
4     <input type="number" step="any" name="Temp9am" placeholder="Temp 9am" required><br>
5     <input type="number" step="any" name="Temp3pm" placeholder="Temp 3pm" required><br>
6     <select name="RainToday" required>
7       <option value="">Rain Today?</option>
8       <option value="1">Yes</option>
9       <option value="0">No</option>
10    </select><br>
11    <input type="number" name="WindGustDir" placeholder="Wind Gust Dir Code" required><br>
12    <input type="number" name="WindDir9am" placeholder="Wind Dir 9am Code" required><br>
13    <input type="number" name="WindDir3pm" placeholder="Wind Dir 3pm Code" required><br>
14    <input type="number" name="year" placeholder="Year" required><br>
15    <input type="number" name="month" placeholder="Month" required><br>
16    <input type="number" name="day" placeholder="Day" required><br>
17    <button type="submit">Predict</button>
18  </form>
19 </body>
20 </html>
```

2. chance.html:

- Now we have to create another html file that is named as chance.html.
- In that file we have to paste the below code.

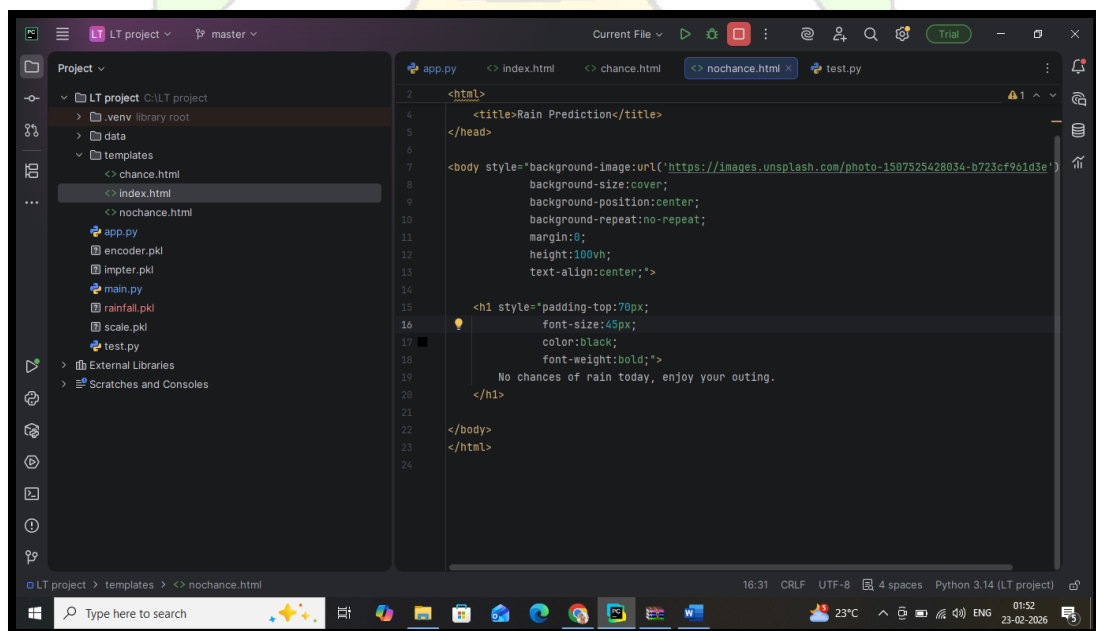
EXPLORATORY ANALYSIS OF RAINFALL DATA IN INDIA FOR AGRICULTURE



```
2 <html>
3 <title>Rain Prediction</title>
4 </head>
5
6 <body style="background-image:url('https://images.unsplash.com/photo-1501594907352-04cda38ebc29');
7   background-size:cover;
8   background-position:center;
9   background-repeat:no-repeat;
10  margin:0;
11  height:100vh;
12  text-align:center;">
13
14   <h1 style="padding-top:80px;
15     font-size:45px;
16     color:black;
17     font-weight:bold;">
18     chances of rain today.
19   </h1>
20
21 </body>
22 </html>
23
24
```

3. Nochance.html:

- Now we have to create another html file that is named as nochance.html.
- In that file we have to paste the below code.

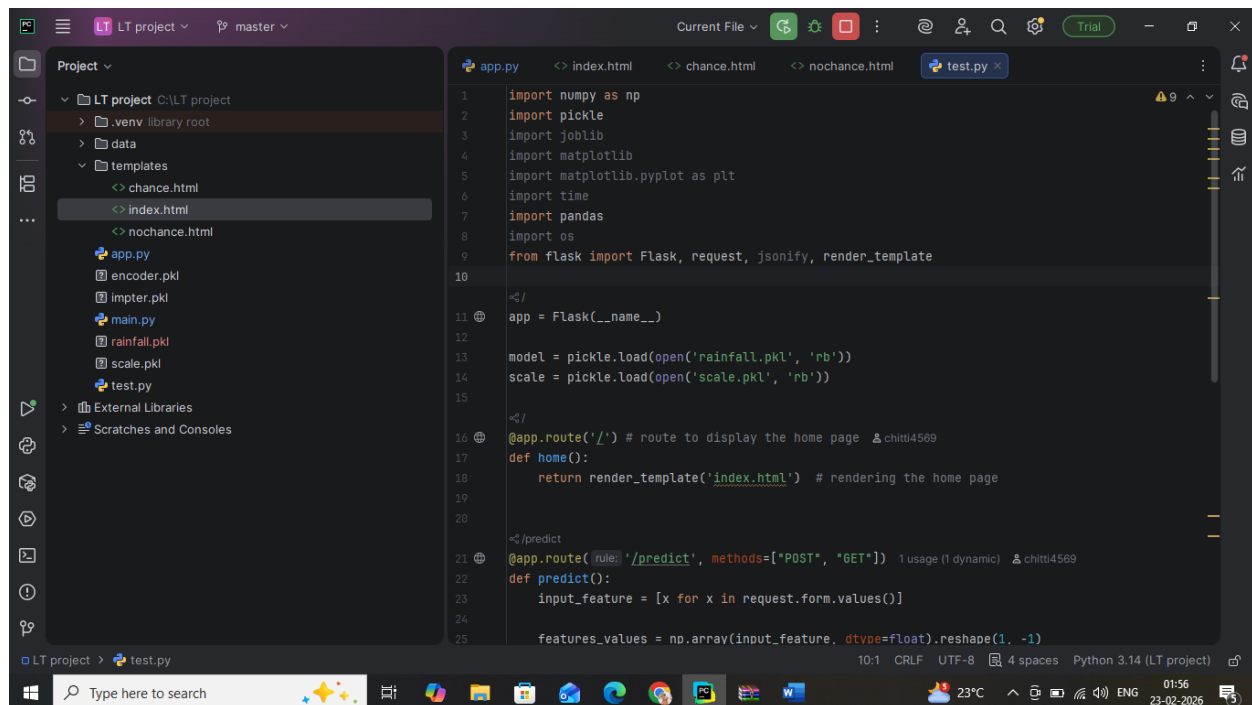


```
2 <html>
3 <title>Rain Prediction</title>
4 </head>
5
6 <body style="background-image:url('https://images.unsplash.com/photo-1507525428034-b723cf96d3e');
7   background-size:cover;
8   background-position:center;
9   background-repeat:no-repeat;
10  margin:0;
11  height:100vh;
12  text-align:center;">
13
14   <h1 style="padding-top:70px;
15     font-size:45px;
16     color:black;
17     font-weight:bold;">
18     No chances of rain today, enjoy your outing.
19   </h1>
20
21 </body>
22 </html>
23
24
```

EXPLORATORY ANALYSIS OF RAINFALL DATA IN INDIA FOR AGRICULTURE

MAIN PYTHON SCRIPT:

- Let us build an test.py flask file which is a web framework written in python for server-side scripting.
- Let's see step by step procedure for building the backend application.
- Now we have to create a python file named as test.py
- Paste the below code on test.py.

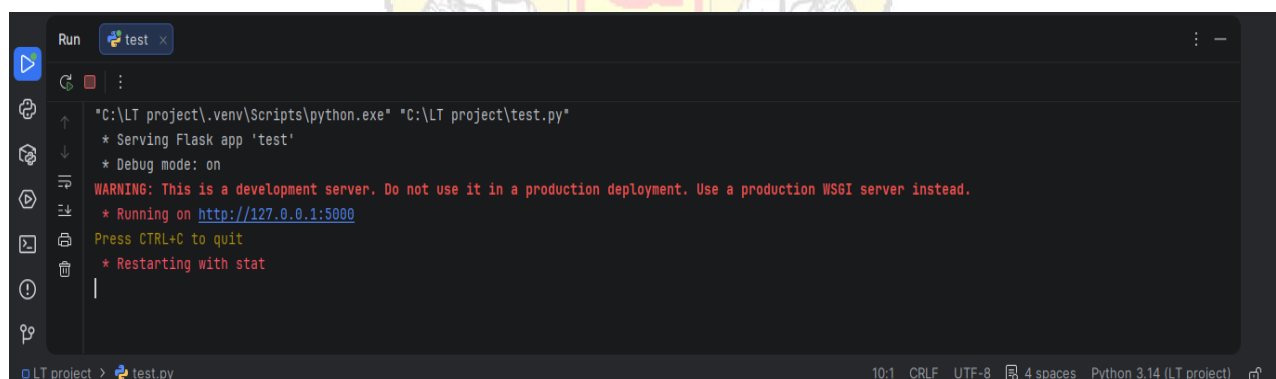


The screenshot shows a code editor with a project named 'LT project'. The file explorer on the left shows the project structure, including a 'templates' folder with 'chance.html', 'index.html', and 'nochance.html', and a 'test.py' file. The main editor window displays the code for 'test.py'.

```
1 import numpy as np
2 import pickle
3 import joblib
4 import matplotlib
5 import matplotlib.pyplot as plt
6 import time
7 import pandas
8 import os
9 from flask import Flask, request, jsonify, render_template
10
11 app = Flask(__name__)
12
13 model = pickle.load(open('rainfall.pkl', 'rb'))
14 scale = pickle.load(open('scale.pkl', 'rb'))
15
16 @app.route('/') # route to display the home page &chitt4569
17 def home():
18     return render_template('index.html') # rendering the home page
19
20
21 @app.route('/predict', methods=['POST', 'GET']) 1 usage (1 dynamic) &chitt4569
22 def predict():
23     input_feature = [x for x in request.form.values()]
24
25     features_values = np.array(input_feature, dtype=float).reshape(1, -1)
```

RUN THE APP:

- Now we have to run the code on test.py.
- Then click on the link on the run bar.



The screenshot shows the 'Run' console output for the 'test.py' file. The output indicates that the Flask app is running on http://127.0.0.1:5000.

```
Run test x
"C:\LT project\.venv\Scripts\python.exe" "C:\LT project\test.py"
* Serving Flask app 'test'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
```

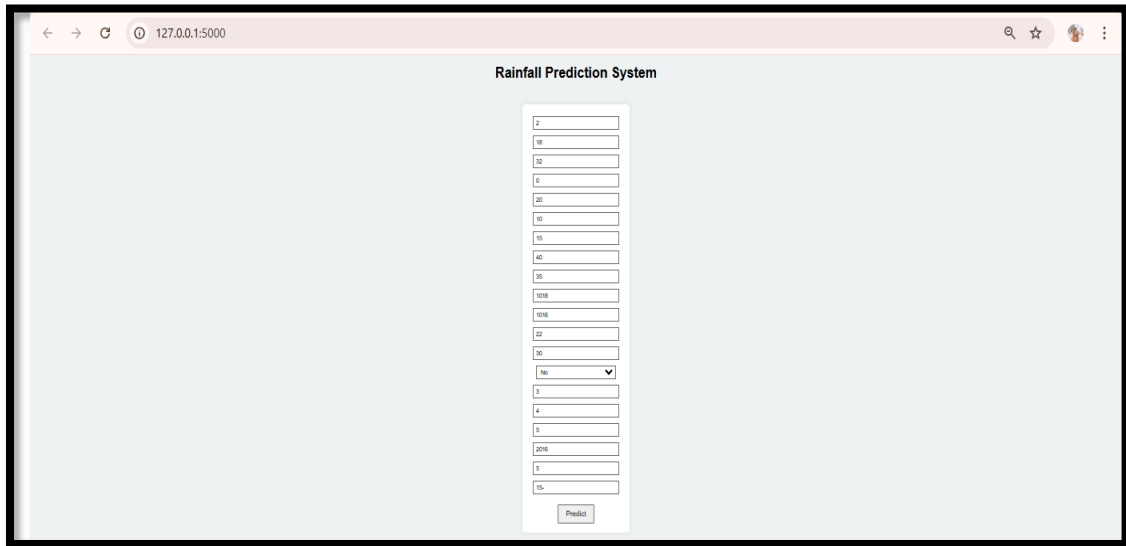
- Navigate to the localhost where you can view your web page, then it will run on local host: 5000

EXPLORATORY ANALYSIS OF RAINFALL DATA IN INDIA FOR AGRICULTURE

OUTPUT:

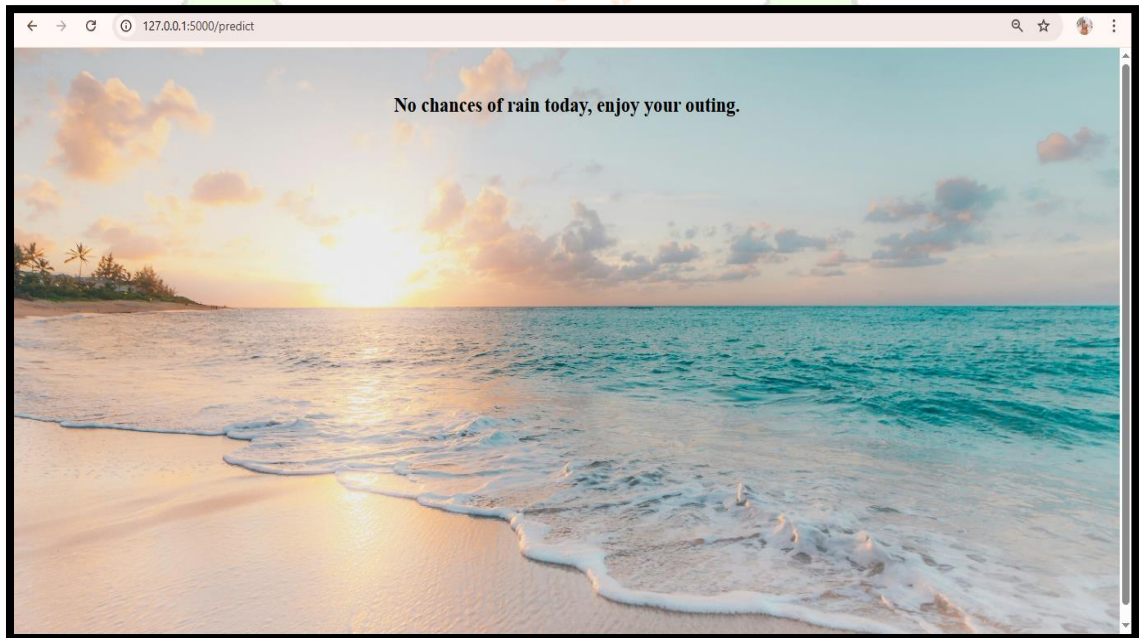
No chance of rain:

- After clicking the link, we have to fill the rain predictions system form.
- For getting the no chance of rain.



The screenshot shows a web browser window with the address bar displaying "127.0.0.1:5000". The page title is "Rainfall Prediction System". The form contains several input fields for numerical data, a dropdown menu, and a "Predict" button. The input fields are arranged in a vertical column and contain the following values from top to bottom: 2, 18, 12, 0, 20, 10, 15, 40, 25, 1018, 1016, 22, 10, 180 (selected in the dropdown), 3, 4, 5, 2016, 5, and 15.

- Now click on predict.
- Now it opens the output.

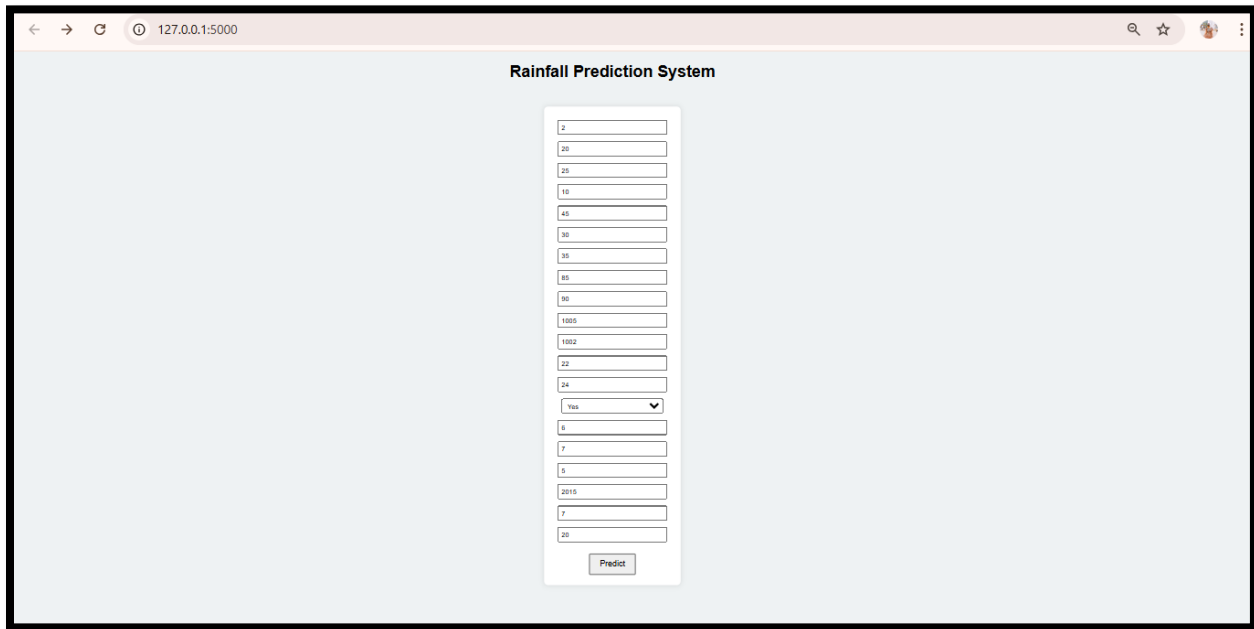


Chances of rain:

- After clicking the link, we have to fill the rain predictions system form.

EXPLORATORY ANALYSIS OF RAINFALL DATA IN INDIA FOR AGRICULTURE

- For getting the chance of rain.



Rainfall Prediction System

2
25
25
10
45
30
35
85
90
1000
1000
22
24
700
6
7
5
2010
7
20

Predict

- Now click on predict.
- Now it opens the output.



ESTD 2001